EXPERIMENT NO. 1

C++ Program: Student Result Management

```
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
    int rollNumber;
    string name;
    float marks[3];
    float total;
    float percentage;
    string result;

public:
    void acceptDetails() {
        cout << "Enter Roll Number : ";
        cin >> rollNumber;

        cin.ignore();

        cout << "Enter Name : ";
        getline(cin, name);

        cout << "Enter marks for 3 subjects : ";

        for (int i = 0; i < 3; i++) {
            cin >> marks[i];
        }
    }

    void calculateResult() {
        total = 0;

        for (int i = 0; i < 3; i++) {
            total += marks[i];
        }

        percentage = total / 3.0;

        if (percentage >= 60)
            result = "First Class";
        else if (percentage >= 50)
            result = "Second Class";
        else if (percentage >= 40)
            result = "Pass";
        else
            result = "Fail";
    }

    void displayDetails() {
        cout << "\n----- Student Details -----" << endl;

        cout << "Roll Number : " << rollNumber << endl;
        cout << "Name : " << name << endl;

        cout << "Marks : ";

        for (int i = 0; i < 3; i++) {
            cout << marks[i] << " ";
        }

        cout << endl;

        cout << "Total Marks : " << total << endl;
        cout << "Percentage : " << percentage << "%" << endl;
        cout << "Result : " << result << endl;
    }
};
```

```
int main() {

    Student s;

    s.acceptDetails();
    s.calculateResult();
    s.displayDetails();

    return 0;
}
```